

## IROZAKI, Japan

WMO: 476660

Lat: 34.6036N

Lon: 138.8428E

Elev: 53

StdP: 100.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 2.9        | 3.9 | -7.8                          | 2.0 | 5.2  | -5.9 | 2.3 | 5.7  | 13.8                  | 8.5  | 13.0 | 8.2  | 6.1                   | 280  | 0.465 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 4.0                    | 29.9            | 26.5 | 29.2 | 26.0 | 28.4 | 25.6 | 27.1                | 29.1 | 26.5 | 28.4 | 26.1 | 27.8 | 3.6                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.5                            | 22.1 | 28.5 | 26.0 | 21.5 | 28.0 | 25.5 | 20.8 | 27.5 | 85.6          | 29.1 | 83.4 | 28.6 | 81.3 | 28.0 | 29.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| 12.4              | 11.1 | 10.0 | DB                         | 1.0  | 31.1               | 1.1 | 1.0                                                | 0.1  | 31.8       | -0.5 | 32.4       | -1.2 | 32.9       | -2.0 | 33.6 | (4) |
|                   |      |      | WB                         | -2.4 | 27.9               | 1.2 | 0.6                                                | -3.2 | 28.3       | -3.9 | 28.7       | -4.5 | 29.0       | -5.4 | 29.4 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 17.0  | 8.2   | 8.4   | 10.9  | 14.9  | 18.6  | 21.2  | 24.9  | 26.6  | 24.1  | 19.8  | 15.3  | 10.8 | (6)  |      |
| (7)  |                                                                   | DBStd       | 6.63  | 2.20  | 2.51  | 2.74  | 2.63  | 1.84  | 1.74  | 2.06  | 1.59  | 2.20  | 2.28  | 2.57  | 2.59 | (7)  |      |
| (8)  |                                                                   | HDD10.0     | 167   | 64    | 57    | 22    | 2     | 0     | 0     | 0     | 0     | 0     | 1     | 21    | (8)  |      |      |
| (9)  |                                                                   | HDD18.3     | 1285  | 314   | 278   | 229   | 105   | 19    | 1     | 0     | 0     | 10    | 95    | 234   | (9)  |      |      |
| (10) |                                                                   | CDD10.0     | 2730  | 8     | 13    | 51    | 148   | 266   | 337   | 461   | 514   | 422   | 305   | 159   | 46   | (10) |      |
| (11) |                                                                   | CDD18.3     | 807   | 0     | 0     | 0     | 2     | 27    | 87    | 203   | 256   | 172   | 57    | 4     | 0    | (11) |      |
| (12) |                                                                   | CDH23.3     | 4721  | 0     | 0     | 0     | 0     | 6     | 111   | 1298  | 2268  | 958   | 79    | 0     | 0    | (12) |      |
| (13) |                                                                   | CDH26.7     | 735   | 0     | 0     | 0     | 0     | 0     | 3     | 186   | 439   | 106   | 1     | 0     | 0    | (13) |      |
| (14) | Wind                                                              | WSAvg       | 4.6   | 5.3   | 5.0   | 5.1   | 4.8   | 4.2   | 4.0   | 3.8   | 3.9   | 4.4   | 4.6   | 4.6   | 5.3  | (14) |      |
| (15) | Precipitation                                                     | PrecAvg     | 1950  | 80    | 107   | 158   | 170   | 189   | 237   | 203   | 154   | 216   | 207   | 141   | 79   | (15) |      |
| (16) |                                                                   | PrecMax     | 2466  | 176   | 213   | 255   | 291   | 327   | 386   | 528   | 449   | 385   | 572   | 407   | 167  | (16) |      |
| (17) |                                                                   | PrecMin     | 1363  | 3     | 27    | 73    | 48    | 84    | 121   | 9     | 2     | 56    | 31    | 26    | 5    | (17) |      |
| (18) |                                                                   | PrecStd     | 299   | 50    | 53    | 51    | 64    | 63    | 72    | 133   | 99    | 92    | 117   | 79    | 43   | (18) |      |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 15.8  | 16.7  | 18.8  | 21.7  | 24.2  | 26.9  | 30.5  | 31.2  | 29.8  | 26.6  | 22.2  | 19.2 | (19) |      |
| (20) |                                                                   | 0.4%        | MCWB  | 13.3  | 14.4  | 14.8  | 17.8  | 20.5  | 23.5  | 26.6  | 27.3  | 26.2  | 22.7  | 18.4  | 17.0 | (20) |      |
| (21) |                                                                   | 2%          | DB    | 14.1  | 15.1  | 17.3  | 20.5  | 23.0  | 25.5  | 29.4  | 30.3  | 28.8  | 25.2  | 21.0  | 17.2 | (21) |      |
| (22) |                                                                   | 2%          | MCWB  | 10.8  | 12.3  | 14.1  | 17.2  | 19.4  | 22.7  | 26.2  | 26.8  | 25.4  | 21.7  | 17.5  | 14.1 | (22) |      |
| (23) |                                                                   | 5%          | DB    | 12.8  | 13.7  | 16.3  | 19.6  | 22.2  | 24.5  | 28.6  | 29.6  | 27.9  | 24.2  | 20.1  | 15.8 | (23) |      |
| (24) |                                                                   | 5%          | MCWB  | 9.2   | 10.6  | 13.5  | 16.4  | 18.8  | 22.2  | 25.7  | 26.4  | 24.9  | 21.0  | 16.9  | 12.2 | (24) |      |
| (25) |                                                                   | 10%         | DB    | 11.6  | 12.4  | 15.4  | 18.6  | 21.4  | 23.8  | 27.7  | 28.9  | 27.0  | 23.2  | 19.2  | 14.6 | (25) |      |
| (26) |                                                                   | 10%         | MCWB  | 7.8   | 9.0   | 12.5  | 15.4  | 18.3  | 21.7  | 25.2  | 26.0  | 24.4  | 20.1  | 15.9  | 10.9 | (26) |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 14.3  | 15.3  | 16.4  | 19.9  | 21.8  | 24.6  | 27.3  | 28.0  | 27.0  | 24.1  | 20.1  | 17.9 | (27) |      |
| (28) |                                                                   | 0.4%        | MCDWB | 15.2  | 16.2  | 17.2  | 20.4  | 23.1  | 25.9  | 29.4  | 30.3  | 28.8  | 25.2  | 21.0  | 18.7 | (28) |      |
| (29) |                                                                   | 2%          | WB    | 11.8  | 13.2  | 15.3  | 18.3  | 20.6  | 23.7  | 26.7  | 27.3  | 26.1  | 22.9  | 18.8  | 14.8 | (29) |      |
| (30) |                                                                   | 2%          | MCDWB | 13.5  | 14.6  | 16.7  | 19.8  | 22.0  | 24.7  | 28.7  | 29.4  | 27.9  | 24.5  | 20.2  | 16.7 | (30) |      |
| (31) |                                                                   | 5%          | WB    | 9.6   | 11.2  | 14.3  | 17.2  | 19.9  | 22.9  | 26.1  | 26.8  | 25.5  | 21.7  | 17.5  | 12.9 | (31) |      |
| (32) |                                                                   | 5%          | MCDWB | 12.2  | 13.2  | 16.0  | 18.8  | 21.2  | 24.0  | 27.9  | 28.7  | 27.2  | 23.7  | 19.5  | 15.2 | (32) |      |
| (33) |                                                                   | 10%         | WB    | 8.1   | 9.4   | 12.9  | 16.2  | 19.1  | 22.1  | 25.6  | 26.4  | 24.9  | 20.5  | 16.4  | 11.3 | (33) |      |
| (34) |                                                                   | 10%         | MCDWB | 11.2  | 12.0  | 15.0  | 18.0  | 20.7  | 23.4  | 27.2  | 28.2  | 26.7  | 22.8  | 18.8  | 14.1 | (34) |      |
| (35) | Mean Daily Temperature Range                                      |             | MDBR  | 4.7   | 5.1   | 5.3   | 5.0   | 4.5   | 3.8   | 3.8   | 4.0   | 3.9   | 3.9   | 4.3   | 4.5  | (35) |      |
| (36) |                                                                   | 5% DB       |       | MCDBR | 5.5   | 6.0   | 5.9   | 5.6   | 5.2   | 4.5   | 4.6   | 4.6   | 4.6   | 5.0   | 5.5  | (36) |      |
| (37) |                                                                   |             |       | MCWBR | 5.3   | 5.8   | 5.5   | 4.5   | 4.0   | 2.9   | 2.6   | 2.7   | 3.0   | 4.0   | 4.8  | 5.3  | (37) |
| (38) |                                                                   |             | 5% WB |       | MCDBR | 5.4   | 5.7   | 5.6   | 4.8   | 4.5   | 3.8   | 4.2   | 4.4   | 4.5   | 4.3  | 4.9  | 5.4  |
| (39) |                                                                   |             |       | MCWBR | 6.0   | 6.4   | 6.1   | 4.6   | 4.0   | 2.8   | 2.6   | 2.8   | 3.2   | 4.2   | 5.4  | 5.9  | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.379 | 0.428 | 0.484 | 0.530 | 0.542 | 0.559 | 0.551 | 0.536 | 0.493 | 0.477 | 0.421 | 0.379 | (40) |      |      |
| (41) |                                                                   | taud        | 2.262 | 2.126 | 1.984 | 1.918 | 1.958 | 1.990 | 2.060 | 2.093 | 2.185 | 2.137 | 2.212 | 2.276 | (41) |      |      |
| (42) |                                                                   | Ebn at Noon | 806   | 801   | 786   | 771   | 766   | 751   | 754   | 757   | 772   | 743   | 750   | 779   | (42) |      |      |
| (43) |                                                                   | Edh at Noon | 107   | 136   | 170   | 190   | 185   | 179   | 166   | 158   | 138   | 134   | 112   | 100   | (43) |      |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 2.69  | 3.26  | 4.14  | 4.78  | 5.20  | 4.56  | 5.25  | 5.36  | 4.17  | 3.17  | 2.58  | 2.41  | (44) |      |      |
| (45) |                                                                   | RadStd      | 0.20  | 0.32  | 0.27  | 0.52  | 0.54  | 0.47  | 0.76  | 0.41  | 0.36  | 0.33  | 0.22  | 0.19  | (45) |      |      |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |         |         |         |     |      |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|-----|------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |     |      |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     | (m) |      |
| (46) | Station Only            | N/A     | N/A    | +1.09   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A | (46) |
| (47) | Regional (36 neighbors) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A | (47) |

Nomenclature: See separate page